ANJALI

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SUMMARY

Final-year AI/ML engineering student with hands-on experience building end-to-end AI-powered applications using Python, FastAPI, React, SQL, and Docker. Experienced with LLM integration, RAG, embeddings, vector search, REST APIs, and databases, with projects spanning GenAI, AI security, and production-oriented systems.

TECHNICAL SKILLS

Languages: Python, SQL, Java
Frontend: React, HTML, CSS
GenAI & LLMs: RAG, Vector Search, Embeddings, Prompt Engineering, LLM Applications, Text-to-SQL
Data Engineering: ETL/ELT pipelines, Feature Engineering, Airflow, Kafka, PostgreSQL, SQLite, Redis, Feast
Backend & APIs: FastAPI, REST APIs, React
ML Frameworks: Scikit-learn, TensorFlow, XGBoost
Infrastructure: Docker, Celery, MLflow, Nginx

WORK EXPERIENCE

The Entrepreneurship Network Jul 2026 – Oct 2026
AI Intern (Remote)

- Built and shipped two GenAI-powered platforms end-to-end: DataSage (LLM-driven Text-to-SQL analytics) and SentinelAI (automated LLM red-teaming) using Python, SQL, FastAPI, React, PostgreSQL, SQLite, Redis, Celery, and Docker.
- Engineered DataSage, a Dockerized Text-to-SQL platform integrating an LLM (Ollama) with a FastAPI + React application layer, translating natural language into validated SQL across four data sources: PostgreSQL, SQLite, document uploads, and demo sources.
- Optimized query execution and schema caching, cutting response latency 42% to under 1.1s for complex multi-table joins; built 10 REST endpoints atop a pluggable PostgreSQL/SQLite data-abstraction layer supporting 200+ requests/sec.
- Standardized deployment with multi-stage Docker builds and Nginx, reducing image size 58% for faster, repeatable rollouts.
- Built SentinelAI, an automated LLM red-teaming platform, implementing custom mutation algorithms and validation logic to detect vulnerabilities such as prompt injection and jailbreaks in model outputs.
- Designed asynchronous data-processing workflows with Celery and Redis to run distributed evaluation and compliance-tracking tasks independently of API request lifecycles, and built a real-time analytics dashboard for automated report generation.
- Worked across both platforms end-to-end: data validation, LLM integration, API design, and containerized deployment applying data quality and testing practices to production-oriented pipelines.

PROJECTS

PaperPilot AI: Local RAG Chatbot & Research Assistant *FastAPI, React, Hugging Face, FAISS*

- Built a FAISS + all-MiniLM-L6-v2 retrieval-augmented generation (RAG) pipeline with embeddings and vector search across 10,000+ text chunks, achieving <15ms retrieval latency and <1s end-to-end query response in a fully local, zero-cost deployment.
- Designed a multi-stage NLP pipeline using BART to extract methodologies, summarize research gaps, and compute readability metrics; optimized local inference via 8-bit quantization, cutting peak memory 53% (8.2GB → 3.8GB) with under 2% ROUGE-L drop.

Argus: Real-Time Fraud Detection MLOps System *Python, XGBoost, Feast, Redis, Kafka, Airflow, MLflow*

- Built a real-time ETL feature pipeline streaming 46 transactional/behavioral features through Kafka into a Redis-backed FastAPI serving layer feeding an XGBoost classifier, achieving <100ms inference latency.
- Implemented automated data-quality and drift monitoring with Airflow and MLflow driven retraining and rollback, cutting detection time to under 5 minutes and deployment lead time by 95%.

AI-Powered Air Quality Monitoring System: ISRO Project *System Engineer & Data Analyst Intern*

- Integrated ESP32-S3 edge sensors via I2C/UART for reliable, non-blocking real-time data ingestion feeding a hybrid LSTM pipeline with physics-validated outputs for AQI forecasting and compliance monitoring.

EDUCATION

Cambridge Institute of Technology 2023 – Present
B.E., Artificial Intelligence & Machine Learning CGPA: 8.00

CERTIFICATIONS

- Introduction to Agent Skills — Anthropic
- GenAI Powered Data Analytics Job Simulation — TATA iQ, Forage
- Computer Graphics — NPTEL